FORBIDDEN SUBGRAPHS IN DIVISOR GRAPHS AND AN ERDŐS DIVISIBILITY PROBLEM

DAMEK DAVIS

Abstract. Erdős asked for the largest size $f(n)$ of a subset of $\{1,\dots,n\}$ with no element dividing two others, and whether $\lim f(n)/n$ is irrational. We show that $f(n) = c_2\, n + o(n)$ for an effectively computable constant c_2, and moreover that the number $q(n)$ of such subsets satisfies $q(n) = \beta_2^{n+o(n)}$ for a computable constant β_2. To prove this, we recast the divisibility constraint as forbidding a certain subgraph in the directed divisor graph on $\{1,\dots,n\}$ and prove a more general result: for any finite family of connected forbidden subgraphs of the divisor graph, both the extremal density and counting rate are effectively computable. The proof uses a theorem of McNew on local statistics of divisor graphs. The irrationality of c_2 remains open.

1. Introduction

Let $f(n)$ denote the largest size of a set $A \subseteq \{1,\dots,n\}$ such that there do not exist distinct $a,b,c \in A$ with $a \mid b$ and $a \mid c$. A question of Erdős (see Guy [Guy04, Problem B24] and Bloom [Blo, Problem 1062]) asks how large $f(n)$ can be and whether $\lim_{n\to\infty} f(n)/n$ is irrational. The interval $\{\lfloor n/3\rfloor+1,\dots,n\}$ already shows that $f(n) \ge \lceil 2n/3\rceil$. Lebensold [Leb77] proved that for sufficiently large n,

$$0.6725\, n \le f(n) \le 0.6736\, n,$$

and OEIS A038372 tabulates the exact extremal values for small n [The].

In this note we prove that for every $\varepsilon > 0$,

$$f(n) = c_2\, n + O_\varepsilon\Big(n \exp\big(-(1-\varepsilon)\sqrt{\log n\, \log\log n}\big)\Big)$$

for an effectively computable constant c_2, and the number of such subsets has exponential growth rate β_2 for a computable constant $\beta_2 \ge 1$ (Corollary 4). The key is to recast the problem in the *divisor graph* on $\{1,\dots,n\}$, whose vertices are joined whenever one divides the other; divisibility gives each edge a natural orientation from divisor to multiple, written $u \to v$ when $u \mid v$. The Erdős condition—no element divides two others—is then equivalent to forbidding the directed *two-fork* $x \to y$, $x \to z$. This makes the problem an instance of a *vertex Turán-type problem* (e.g., [JT10]): given a host graph H and a forbidden subgraph F, find the largest vertex subset $U \subseteq V(H)$ whose induced subgraph contains no copy of F. (The classical theorem of Turán [Tur41] is the edge-maximization analogue.)

Our main result (Theorem 1) shows that for any finite family of connected forbidden subgraphs of the divisor graph, the largest avoiding subset of $\{1,\dots,n\}$ has size $c\,n + o(n)$ for an effectively computable constant c (Corollary 3). In addition, the number of avoiding subsets has exponential growth rate β for an effectively computable constant $\beta \ge 1$. The forbidden subgraphs may be directed or undirected; for the Erdős problem the orientation is essential, since the undirected two-fork would also forbid the reverse pattern—no element is a common multiple of two others—which is a different condition. The Erdős two-fork is one instance (Corollary 4); further applications to r-fork avoidance and k-chain avoidance are given in Remark 5.

The main tool is a theorem of McNew [McN21] on local statistics of divisor graphs, which expresses certain additive and multiplicative divisor-graph statistics as explicit convergent series. Theorem 1

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identifies a broad class of properties—those that are downward closed, decompose over coprime components, and are preserved by integer dilation—for which we use McNew's theorem to obtain both computable density and counting estimates. For the Erdős two-fork case, the numerical estimates in Section 5 improve Lebensold's lower bound to $c_2 \geq 0.6729\ldots$, narrowing the gap with his upper bound $c_2 \leq 0.6736$ to approximately 0.0007. We also estimate $1.729\ldots \leq \beta_2 \leq 1.874\ldots$.

McNew [McN21] also uses his theorem to study other problems on the divisor graph. The most-studied case is the two-chain $a \mid b$: sets avoiding it are called *primitive*, and their extremal density is 1/2. Determining the counting rate is more involved: Angelo [Ang18] proved that the number of primitive subsets of $\{1, \ldots, n\}$ has an exponential growth rate, and McNew [McN21, Theorem 2] obtained this rate with an explicit error term.

2. Main results

This section states the general theorem (Theorem 1), which gives both an extremal density and a counting result, derives its forbidden-subgraph corollary (Corollary 3), and specializes to the Erdős problem (Corollary 4). We begin with the notation needed for the constants that appear in these results.

For any family $\mathcal{P}$ of finite subsets of $\mathbb{N}$ and any finite set $T \subset \mathbb{N}$, write

$$\Phi_{\mathcal{P}}(T) := \max\{|B| : B \subseteq T,\ B \in \mathcal{P}\}, \qquad Q_{\mathcal{P}}(T) := \#\{B \subseteq T : B \in \mathcal{P}\}.$$

Let $P^+(d)$ denote the largest prime factor of d, with $P^+(1) = 1$. For any function Ψ defined on finite subsets of $\mathbb{N}$ for which the following series converges absolutely, define

$$C_\Psi := \sum_{i=1}^{\infty} \left(\prod_{p \leq i} \frac{p-1}{p} \right) \sum_{\substack{d \geq 1 \\ P^+(d) \leq i}} \sum_{t \in [id, (i+1)d)} \frac{\Psi(\{d, \ldots, t\}) - \Psi(\{d+1, \ldots, t\})}{t(t+1)}.$$

The coefficient weights in this series sum to 1 (as we later see in (1)), so C_Ψ converges absolutely whenever the differences $\Psi(\{d, \ldots, t\}) - \Psi(\{d+1, \ldots, t\})$ are uniformly bounded. In particular, $C_{\Phi_{\mathcal{P}}}$ and $C_{\log Q_{\mathcal{P}}}$ are well-defined for any family $\mathcal{P}$ satisfying the hypotheses of the theorem below.

Theorem 1. *Let $\mathcal{P}$ be a family of finite subsets of $\mathbb{N}$, called admissible sets, and assume $\emptyset \in \mathcal{P}$. Assume further:*

(1) (downward closed) if $B \in \mathcal{P}$ and $B' \subseteq B$, then $B' \in \mathcal{P}$;
(2) (decomposes over coprime components) if $T = T_1 \sqcup T_2$ is finite and there is no divisibility relation between an element of T_1 and an element of T_2, then for every $B \subseteq T$ we have $B \in \mathcal{P}$ if and only if $B \cap T_i \in \mathcal{P}$ for $i = 1, 2$;
(3) (preserved by integer dilation) if $m \in \mathbb{N}$ and B is finite, then $B \in \mathcal{P}$ if and only if $mB \in \mathcal{P}$, where $mB := \{mb : b \in B\}$.

Write $f_{\mathcal{P}}(n) := \Phi_{\mathcal{P}}(\{1, \ldots, n\})$, $q_{\mathcal{P}}(n) := Q_{\mathcal{P}}(\{1, \ldots, n\})$, $c_{\mathcal{P}} := C_{\Phi_{\mathcal{P}}}$, and $\beta_{\mathcal{P}} := \exp(C_{\log Q_{\mathcal{P}}})$. Then for every $\varepsilon > 0$:

(a) (Extremal density.)

$$f_{\mathcal{P}}(n) = c_{\mathcal{P}}\, n + O_\varepsilon\Big(n \exp\big(-(1-\varepsilon)\sqrt{\log n \,\log\log n}\big)\Big).$$

(b) (Counting.)

$$\log q_{\mathcal{P}}(n) = n \log \beta_{\mathcal{P}} + O_\varepsilon\Big(n \exp\big(-(1-\varepsilon)\sqrt{\log n \,\log\log n}\big)\Big).$$

In particular, $\lim_{n\to\infty} q_{\mathcal{P}}(n)^{1/n} = \beta_{\mathcal{P}}$.

If in addition membership in $\mathcal{P}$ is decidable for finite sets, then $c_{\mathcal{P}}$ and $\beta_{\mathcal{P}}$ are effectively computable.

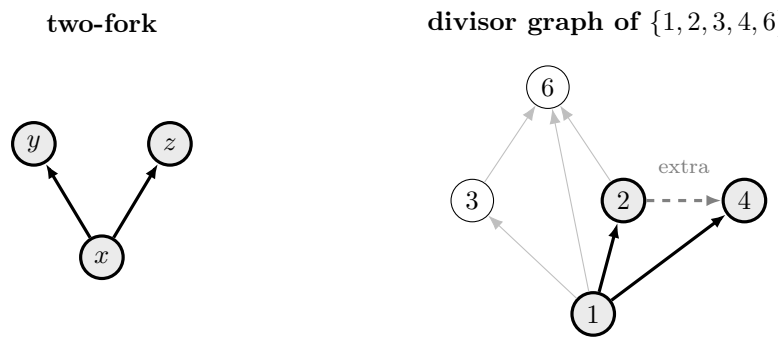

FIGURE 1. Non-induced subgraph containment. **Left:** the two-fork. **Right:** the divisor graph of $\{1,2,3,4,6\}$; the shaded vertices $\{1,2,4\}$ realize the two-fork via the bold arrows $1 \to 2$ and $1 \to 4$. The dashed arrow $2 \to 4$ is an extra divisibility among the matched vertices; the copy need not be induced.

Remark 2 (Partition function)**.** The same argument shows that for any fixed $z > 0$, the partition function $Z_{\mathcal{P}}(T, z) := \sum_{B \subseteq T,\, B \in \mathcal{P}} z^{|B|}$ satisfies

$$\log Z_{\mathcal{P}}(\{1, \dots, n\}, z) = \kappa_{\mathcal{P}}(\log z)\, n + O_{\varepsilon}\Big(\log(1+z)\, n \exp\big(-(1-\varepsilon)\sqrt{\log n \, \log\log n}\big)\Big)$$

for an effectively computable constant $\kappa_{\mathcal{P}}(\log z)$. The case $z = 1$ is part (b) of the theorem. It is straightforward to show that the "pressure" $t \mapsto \kappa_{\mathcal{P}}(t)$ is convex and 1-Lipschitz [CGT09], and $\kappa_{\mathcal{P}}(t)/t \to c_{\mathcal{P}}$ as $t \to \infty$.

We now describe the family of admissible sets to which we apply Theorem 1. The *divisor graph* of a finite set $S \subset \mathbb{N}$ has vertex set S with an edge between u and v whenever $u \neq v$ and one divides the other; we orient each edge from divisor to multiple, writing $u \to v$ when $u \mid v$. Given a collection $\mathcal{F}$ of connected graphs, each directed or undirected—here connected refers to the underlying undirected graph, regardless of orientation—say that a finite set S is *$\mathcal{F}$-free* if its divisor graph contains no copy of any member of $\mathcal{F}$: containment of a directed member means containment as a directed subgraph of the oriented divisor graph, while containment of an undirected member means containment as a subgraph of the underlying undirected graph. Write $\mathcal{P}(\mathcal{F})$ for the family of all $\mathcal{F}$-free sets. We do not require the copy to be induced, so extra divisibility arrows among the chosen vertices are allowed.[1] For example, $\{1, 2, 4\}$ contains the two-fork $x \to y$, $x \to z$: the arrows $1 \to 2$ and $1 \to 4$ realize it, while the extra arrow $2 \to 4$ is harmless. Figure 1 illustrates this distinction, and Figure 2 shows four representative connected forbidden subgraphs. The family $\mathcal{P}(\mathcal{F})$ satisfies the three axioms of Theorem 1 (proof in Section 4), giving the following corollary.

Corollary 3. *Let $\mathcal{F}$ be a finite family of connected[2] forbidden subgraphs (directed or undirected) of divisor graphs, and let $f_{\mathcal{F}}(n)$ and $q_{\mathcal{F}}(n)$ denote the maximum size, and the number, of $\mathcal{F}$-free subsets of $\{1, \dots, n\}$. Applying Theorem 1 to $\mathcal{P}(\mathcal{F})$, there exist effectively computable constants $c_{\mathcal{F}}$ and $\beta_{\mathcal{F}} \geq 1$ such that, for every $\varepsilon > 0$,*

$$f_{\mathcal{F}}(n) = c_{\mathcal{F}}\, n + O_{\varepsilon}\big(n \exp\big(-(1-\varepsilon)\sqrt{\log n \, \log\log n}\big)\big),$$
$$\log q_{\mathcal{F}}(n) = n \log \beta_{\mathcal{F}} + O_{\varepsilon}\big(n \exp\big(-(1-\varepsilon)\sqrt{\log n \, \log\log n}\big)\big).$$

[1] If one instead requires induced containment, the three axioms still hold and Theorem 1 applies equally. Non-induced avoidance is more restrictive and is the natural choice for the Erdős problem, since "no element divides two others" does not depend on the relationship between those two others.

[2] Connectedness is essential: a disconnected forbidden subgraph can straddle two connected components of the divisor graph, so the componentwise axiom may fail.

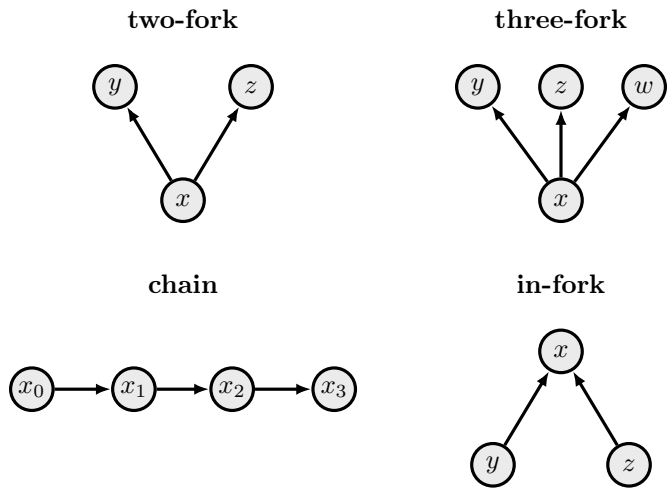

FIGURE 2. Representative connected forbidden subgraphs of divisor graphs. Arrows point from divisor to multiple.

Corollary 3 is not the only source of examples for Theorem 1. The collection $\mathcal{F}$ may also be infinite, or may mix directed and undirected patterns; the only requirement for Theorem 1 is that the resulting family $\mathcal{P}(\mathcal{F})$ be decidable and satisfy the three axioms. (Finiteness of $\mathcal{F}$ in Corollary 3 guarantees decidability automatically.) For instance, requiring the divisor graph of the chosen subset to be a forest—equivalently, forbidding all cycles as undirected subgraphs—satisfies the three axioms, so Theorem 1 applies. This is a genuinely infinite family of forbidden subgraphs: for every $k \geq 3$, taking k distinct primes $p_1, \dots, p_k$ and the products p_1p_2, p_2p_3, $\dots$, p_kp_1 yields a set of $2k$ integers whose divisor graph is exactly C_{2k} with no shorter cycle, so no finite subfamily suffices.

Specializing the corollary to the two-fork $x \mid y$, $x \mid z$ recovers the Erdős problem:

Corollary 4. *Let $f(n)$ and $q(n)$ denote the maximum size, and the number, of subsets of $\{1, \dots, n\}$ containing no distinct x, y, z with $x \mid y$ and $x \mid z$. Then there exist effectively computable constants c_2 and $\beta_2 \geq 1$ such that, for every $\varepsilon > 0$,*

$$f(n) = c_2\, n + O_\varepsilon\Big(n \exp\big(-(1-\varepsilon)\sqrt{\log n \,\log\log n}\big)\Big)\,,$$

$$\log q(n) = n \log \beta_2 + O_\varepsilon\Big(n \exp\big(-(1-\varepsilon)\sqrt{\log n \,\log\log n}\big)\Big)\,.$$

In particular, $\lim_{n\to\infty} q(n)^{1/n} = \beta_2$.

The question of whether c_2 is irrational, raised by Erdős (see [Guy04, Problem B24]), remains open.

Remark 5 (Forks, in-forks, and chains)**.** Corollary 3 applies equally to other connected forbidden subgraphs. For each $r \geq 2$, the r-fork—forbidding any element from dividing r others—is connected, so the corollary gives computable density c_r and counting rate. The interval $(\lfloor n/(r+1)\rfloor, n]$ gives $c_r \geq r/(r+1)$ and, since all of its subsets are admissible, $\beta_r \geq 2^{r/(r+1)}$. For $r \geq 3$, earlier bounds on c_r were obtained by Vijay [Vij06] and Hegarty [Heg06].

For each $k \geq 2$, the directed path $x_1 \to x_2 \to \cdots \to x_k$ on k vertices is connected, so the corollary gives a computable density γ_k and counting rate δ_k for k-chain-free subsets of $\{1, \dots, n\}$. The interval $(\lfloor n/2^{k-1}\rfloor, n]$ is k-chain-free, giving $\gamma_k \geq 1 - 2^{1-k}$ and $\delta_k \geq 2^{1-2^{1-k}}$. For $k = 2$, $\gamma_2 = 1/2$: a 2-chain-free set is an antichain, and the $\lceil n/2\rceil$ chains $\{q, 2q, 4q, \dots\} \cap [1, n]$ for odd $q \leq n$ show this is tight.

Dually, the r-in-fork $y_1 \to x, \ldots, y_r \to x$ forbids any element from being a multiple of r others. This is the problem dual to the r-fork, also posed by Erdős [Guy04, Problem B24]. The in-fork is connected and directed, so Corollary 3 gives a computable density c_r^* and counting rate for each $r \geq 2$. The same interval $(\lfloor n/(r+1) \rfloor, n]$ is also r-in-fork-free, giving $c_r^* \geq r/(r+1)$. Following the same techniques as in Section 5, we obtain $0.7195 \leq c_2^* \leq 0.788$ and $1.82 \leq \beta_2^* \leq 1.91$, confirming that the in-fork and two-fork problems have different extremal densities.

3. Proof of Theorem 1

We deduce Theorem 1 from the following result of McNew. For integers $1 \leq a \leq n$, let $D[a,n]$ be the divisor graph on the vertex set $\{a, a+1, \ldots, n\}$, with an edge between u and v when $u \mid v$ or $v \mid u$.

Theorem 6 (McNew [McN21, Theorem 3 and footnote 1])**.** *Fix $\varepsilon > 0$. Let $h(a,n)$ be defined for $1 \leq a \leq n$, assume $|h(a,n)| \leq M$, and suppose that $h(a,n)$ depends only on the rooted connected component of a in $D[a,n]$, that is, on the connected component together with the distinguished vertex a, up to linear scaling isomorphism.*[3] *Then there exists a constant c_h such that*

$$\sum_{a=1}^{n} h(a,n) = c_h\, n + O_\varepsilon\Big(Mn \exp\big(-(1-\varepsilon)\sqrt{\log n \,\log\log n}\big)\Big).$$

Moreover, if $P^+(d)$ denotes the largest prime factor of d, with the convention $P^+(1) = 1$, then

$$c_h = \sum_{i=1}^{\infty} \left(\prod_{p \leq i} \frac{p-1}{p}\right) \sum_{\substack{d \geq 1 \\ P^+(d) \leq i}} \sum_{t \in [id, (i+1)d)} \frac{h(d,t)}{t(t+1)}.$$

The proof verifies McNew's hypotheses for suitable local functions and then recovers the stated asymptotics by telescoping.

Proof of Theorem 1. Part (a)*: extremal density.* Write $F_{\mathcal{P}}(a,n) := \Phi_{\mathcal{P}}(\{a, \ldots, n\})$ and $g_{\mathcal{P}}(a,n) := F_{\mathcal{P}}(a,n) - F_{\mathcal{P}}(a+1,n)$, with $F_{\mathcal{P}}(n+1,n) = 0$.

Since $\{a+1, \ldots, n\} \subseteq \{a, \ldots, n\}$, we have $F_{\mathcal{P}}(a,n) \geq F_{\mathcal{P}}(a+1,n)$. By downward closure, if $B \subseteq \{a, \ldots, n\}$ is admissible, then $B \setminus \{a\} \subseteq \{a+1, \ldots, n\}$ is admissible, so $F_{\mathcal{P}}(a+1,n) \geq F_{\mathcal{P}}(a,n) - 1$. Hence $0 \leq g_{\mathcal{P}}(a,n) \leq 1$.

Next we show that $g_{\mathcal{P}}(a,n)$ depends only on the rooted connected component of a in $D[a,n]$. If $T = T_1 \sqcup T_2$ and there is no divisibility relation between an element of T_1 and an element of T_2, then the second axiom says that a subset $B \subseteq T$ is admissible if and only if $B \cap T_i \in \mathcal{P}$ for $i = 1, 2$. Thus admissible subsets of T are exactly the unions $B_1 \sqcup B_2$ with $B_i \in \mathcal{P}$, so $\Phi_{\mathcal{P}}(T) = \Phi_{\mathcal{P}}(T_1) + \Phi_{\mathcal{P}}(T_2)$. Now let $C(a,n)$ be the connected component of a in $D[a,n]$, and let

$$R(a,n) := \{a, \ldots, n\} \setminus C(a,n).$$

Note that $\{a, \ldots, n\} = C(a,n) \sqcup R(a,n)$ and $\{a+1, \ldots, n\} = (C(a,n) \setminus \{a\}) \sqcup R(a,n)$, with no divisor-graph edges between $C(a,n)$ and $R(a,n)$. Hence

$$F_{\mathcal{P}}(a,n) = \Phi_{\mathcal{P}}(C(a,n)) + \Phi_{\mathcal{P}}(R(a,n)), \qquad F_{\mathcal{P}}(a+1,n) = \Phi_{\mathcal{P}}(C(a,n) \setminus \{a\}) + \Phi_{\mathcal{P}}(R(a,n)),$$

and subtraction gives

$$g_{\mathcal{P}}(a,n) = \Phi_{\mathcal{P}}(C(a,n)) - \Phi_{\mathcal{P}}(C(a,n) \setminus \{a\}).$$

Thus $g_{\mathcal{P}}(a,n)$ depends only on the rooted connected component of a in $D[a,n]$.

[3] McNew states the theorem for rooted graph isomorphism; the accompanying footnote observes that one may restrict the definition to linear isomorphisms without affecting the proof.

If $C(a,n)$ and $C(b,m)$ are linearly isomorphic, choose $\lambda = u/v > 0$ with $u, v \in \mathbb{N}$ and $u\,C(a,n) = v\,C(b,m)$. By scale invariance,

$$\Phi_{\mathcal{P}}(C(a,n)) = \Phi_{\mathcal{P}}(C(b,m)), \qquad \Phi_{\mathcal{P}}(C(a,n) \setminus \{a\}) = \Phi_{\mathcal{P}}(C(b,m) \setminus \{b\}).$$

Hence $g_{\mathcal{P}}(a,n) = g_{\mathcal{P}}(b,m)$. Therefore $g_{\mathcal{P}}$ satisfies Theorem 6 with $M = 1$, and so

$$\sum_{a=1}^{n} g_{\mathcal{P}}(a,n) = c_{\mathcal{P}}\, n + O_{\varepsilon}\Big(n \exp\big(-(1-\varepsilon)\sqrt{\log n \,\log\log n}\big)\Big),$$

Comparing McNew's formula for c_h with the definition of C_{Ψ} shows that $c_{\mathcal{P}} = C_{\Phi_{\mathcal{P}}}$, as claimed. Telescoping gives

$$f_{\mathcal{P}}(n) = F_{\mathcal{P}}(1,n) = \sum_{a=1}^{n} \big(F_{\mathcal{P}}(a,n) - F_{\mathcal{P}}(a+1,n)\big) = \sum_{a=1}^{n} g_{\mathcal{P}}(a,n),$$

which yields the stated asymptotic.

It remains to show effective computability. Since $0 \le g_{\mathcal{P}}(d,t) \le 1$, every term in the series $C_{\Phi_{\mathcal{P}}}$ is nonnegative. Write $w_{i,d,t} := \big(\prod_{p \le i} \frac{p-1}{p}\big) \frac{1}{t(t+1)}$ for the coefficient weight of the (i,d,t) term, so that $C_{\Phi_{\mathcal{P}}} = \sum w_{i,d,t}\, g_{\mathcal{P}}(d,t)$. By partial fractions and the Euler product for smooth numbers,

$$\sum_{t \in [id,(i+1)d)} \frac{1}{t(t+1)} = \frac{1}{i(i+1)d} \qquad \text{and} \qquad \sum_{\substack{d \ge 1 \\ P^+(d) \le i}} \frac{1}{d} = \prod_{p \le i} \left(1 - \frac{1}{p}\right)^{-1}, \tag{1}$$

so the total weight is $\sum w_{i,d,t} = \sum_{i=1}^{\infty} 1/(i(i+1)) = 1$. In particular, the series $C_{\Phi_{\mathcal{P}}}$ converges absolutely. If membership in $\mathcal{P}$ is decidable for finite sets, then $g_{\mathcal{P}}(d,t)$ is computable by exhaustive search. Enumerate the triples (i,d,t) in any computable order, and let

$$S_N := \sum_{n \le N} w_{i_n,d_n,t_n}\, g_{\mathcal{P}}(d_n,t_n), \qquad W_N := \sum_{n \le N} w_{i_n,d_n,t_n}.$$

Since all terms are nonnegative and $\sum w_{i,d,t} = 1$, we have

$$0 \le c_{\mathcal{P}} - S_N \le 1 - W_N.$$

Thus once $1 - W_N < \delta$, the partial sum S_N approximates $c_{\mathcal{P}}$ to within δ.

Part (b)*: counting.* The argument is the multiplicative analogue of part (a). Write $Q_{\mathcal{P}}(a,n) := Q_{\mathcal{P}}(\{a,\dots,n\})$, with $Q_{\mathcal{P}}(n+1,n) = 1$. If $T = T_1 \sqcup T_2$ with no divisibility between T_1 and T_2, the componentwise axiom says $B \subseteq T$ is admissible if and only if $B \cap T_i$ is admissible for $i = 1,2$, so admissible subsets of T are exactly the unions $B_1 \sqcup B_2$ with each $B_i \subseteq T_i$ admissible. Hence $Q_{\mathcal{P}}(T) = Q_{\mathcal{P}}(T_1)\, Q_{\mathcal{P}}(T_2)$. Define

$$h_{\mathcal{P}}(a,n) := \log \frac{Q_{\mathcal{P}}(a,n)}{Q_{\mathcal{P}}(a+1,n)}.$$

The multiplicative decomposition over $C(a,n) \sqcup R(a,n)$ cancels $Q_{\mathcal{P}}(R(a,n))$ in numerator and denominator, so $h_{\mathcal{P}}(a,n)$ depends only on the rooted component of a. Every admissible subset of $\{a+1,\dots,n\}$ is also an admissible subset of $\{a,\dots,n\}$, so $Q_{\mathcal{P}}(a,n) \ge Q_{\mathcal{P}}(a+1,n)$. Conversely, every admissible $B \subseteq \{a,\dots,n\}$ either omits a or contains it, and removing a preserves admissibility by downward closure, so $Q_{\mathcal{P}}(a,n) \le 2\,Q_{\mathcal{P}}(a+1,n)$. Hence $0 \le h_{\mathcal{P}}(a,n) \le \log 2$. Scale invariance preserves both counts, so $h_{\mathcal{P}}$ is invariant under linear isomorphism of rooted components. Hence $h_{\mathcal{P}}$ satisfies Theorem 6 with $M = \log 2$. The ratios telescope: $\log q_{\mathcal{P}}(n) = \sum_{a=1}^{n} h_{\mathcal{P}}(a,n)$, and McNew gives $\sum h_{\mathcal{P}}(a,n) = C_{\log Q_{\mathcal{P}}}\, n + O_{\varepsilon}(\cdots)$, so $\beta_{\mathcal{P}} = \exp(C_{\log Q_{\mathcal{P}}})$. Effective computability follows from the same weight argument as in part (a), with $\log 2$ replacing 1. $\square$

4. Connected forbidden subgraphs

This section verifies the hypotheses of Theorem 1 for forbidden-subgraph avoidance and proves Corollary 3.

Fix a finite family $\mathcal{F}$ of connected forbidden subgraphs (directed or undirected). Let $\mathcal{P}(\mathcal{F})$ be the family of finite sets $B \subset \mathbb{N}$ whose divisor graph contains no copy of any member of $\mathcal{F}$.

Proposition 7. *The family $\mathcal{P}(\mathcal{F})$ satisfies the hypotheses of Theorem 1.*

Proof. Clearly $\emptyset \in \mathcal{P}(\mathcal{F})$. The downward-closed axiom is immediate: passing to a subset cannot create a forbidden subgraph. For scale invariance, note that the axiom only asks about integer dilations. If $m \in \mathbb{N}$, then multiplication by m preserves divisibility, so a finite set B contains a forbidden subgraph if and only if mB does. Hence $B \in \mathcal{P}(\mathcal{F})$ if and only if $mB \in \mathcal{P}(\mathcal{F})$.

For the componentwise axiom, let $T = T_1 \sqcup T_2$ with no divisibility relation between T_1 and T_2. If $B \subseteq T$ contains a copy of some forbidden subgraph in $\mathcal{F}$, let $S \subseteq B$ be the set of vertices of that copy. Since the forbidden subgraph is connected after forgetting directions, its image on S is connected in the undirected divisor graph, so S lies entirely in T_1 or entirely in T_2. Therefore $B \in \mathcal{P}(\mathcal{F})$ if and only if both $B \cap T_1$ and $B \cap T_2$ lie in $\mathcal{P}(\mathcal{F})$. □

Proof of Corollary 3. Apply Theorem 1 with $\mathcal{P} = \mathcal{P}(\mathcal{F})$, using Proposition 7. Effective computability follows because $\mathcal{F}$ is finite, so membership in $\mathcal{P}(\mathcal{F})$ is decidable for finite sets. □

5. Numerical estimates

Both constants c_2 and β_2 are computed from the same series C_Ψ, with $\Psi = \Phi$ for the density and $\Psi = \log Q$ for the counting rate. Write $g(d,t) := \Psi(\{d,\dots,t\}) - \Psi(\{d+1,\dots,t\})$ for the local increment. In both cases $g(d,t) \geq 0$, so every term in the series is nonnegative and any finite partial sum is a lower bound on C_Ψ. The coefficient weights $w_{i,d,t} := \left(\prod_{p\leq i} \frac{p-1}{p}\right) \frac{1}{t(t+1)}$ sum to 1 (see (1)), and the maximum contribution of a fixed (i,d) block is

$$w_{i,d} \;:=\; \sum_{t\in[id,(i+1)d)} w_{i,d,t} \;=\; \frac{1}{i(i+1)d} \prod_{p\leq i} \frac{p-1}{p},$$

so small d and especially small i carry most of the mass. We truncate to triples satisfying $d\, i^\alpha \leq B$ for parameters $\alpha \geq 1$ and $B \geq 1$. Let S be the resulting partial sum and $W = \sum w_{i,d,t}$ the retained coefficient mass. Since $g(d,t) \leq M$ for a known bound M (with $M = 1$ for c_2 and $M = \log 2$ for β_2), the omitted tail is at most $M(1-W)$, giving

$$S \;\leq\; C_\Psi \;\leq\; S + M(1-W).$$

Both a lower and an upper bound thus follow from a single truncation. Each local increment $g(d,t)$ is computed exactly for each rooted-component block: for c_2, this requires solving a finite 0-1 optimization; for β_2, it requires counting all admissible subsets by dynamic programming.

5.1. A numerical estimate for c_2.

For the density, $g_\Phi(d,t) := \Phi(\{d,\dots,t\}) - \Phi(\{d+1,\dots,t\})$ with $M = 1$. We evaluated $g_\Phi(d,t)$ exactly using the CP-SAT solver in Google OR-Tools. With $\alpha = 10$ and $B = 10^{13}$, the truncation retains 32660 rooted-component blocks and gives

$$c_2 \;\geq\; 0.6729.$$

Combined with Lebensold's upper bound $c_2 \leq 0.6736$ from [Leb77], this leaves a gap of approximately 7×10^{-4}. An analogous upper bound from our series would require the omitted tail to be smaller than Lebensold's bound, but the tail decays as $1/(I+1)$ for an $i \leq I$ truncation, so reaching this level would require evaluating far more blocks.

5.2. **A numerical estimate for β_2.** For the counting rate, $h_Q(d,t) := \log Q(\{d,\dots,t\}) - \log Q(\{d+1,\dots,t\})$ with $M = \log 2$. We evaluated $h_Q(d,t)$ exactly by enumerating all subsets of each rooted-component block via dynamic programming, counting how many are two-fork-free for both the block and its root-deleted version. With $\alpha = 10$ and $B = 10^{10}$, the truncation retains 1143 blocks. The resulting bounds are

$$1.729\ldots \le \beta_2 \le 1.874\ldots.$$

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Department of Statistics and Data Science, The Wharton School, University of Pennsylvania
Email address: `damek@wharton.upenn.edu`